Quantum field theory cheat sheet
– canonical quantisation –

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Abstract

From David Tong Quantum Field Theory Part III Mathematical Tripos course and Peskin Schroeder "An introduction to Quantum Field Theory"

Dyson formula, Schrödinger, Heisenberg, Interaction picture	
(D) **Dyson formula, time-ordering operator**	The solution to the equation for the operator $U_H(t, t_0)$ $$\frac{d}{dt}U_H(t) = -iH(t)U_H(t, t_0)$$ where $H(t)$ is a time-dependent operator, is given by the Dyson formula: $$U_H(t, t_0) = T\left[\exp\left(-i\int_{t_0}^{t} H(s)ds\right)\right]$$ where T is the **time ordering operator** ($f = 1$ for fermions) $$T[H(t_1)H(t_2)] = \begin{cases} H(t_1)H(t_2) \text{ if } t_1 \geq t_2 \\ (-1)^f H(t_2)H(t_1) \text{ if } t_1 < t_2 \end{cases}$$
(D) **Schrödinger, Heisenberg, Interaction picture**	Let the Hamiltonian be $H = H_0 + H_I$ where typically H_0 describes the interaction-free laws and H_I describes the interactions between fields. In the **Schrödinger picture**: The operators O_S and the bases of the Hilbert spaces are time-independent. The states $\lvert\phi(t)\rangle_S = U(t, t_0)\lvert\phi(t_0)\rangle_S$ are time-dependent. In the **Heisenberg picture**: The operators $O_H(t) = U^\dagger(t, t_0)O_H(t_0)U(t, t_0)$ and the bases $\lvert i(t)\rangle_H = U^\dagger(t, t_0)\lvert i(t_0)\rangle_H$ of the Hilbert spaces are time-dependent, and $\frac{dO_H}{dt} = i[H, O_H]$. The states $\lvert\phi\rangle_H$ are time-dependent. In the **interaction picture**: The operators $O_I(t) = U^\dagger_{H_0}(t, t_0)O_I(t_0)U_{H_0}(t, t_0)$ and the bases $\lvert i(t)\rangle_I = U^\dagger_{H_0}(t, t_0)\lvert i(t_0)\rangle_I$ of the Hilbert spaces are time-dependent, and $\frac{dO_H}{dt} = i[H, O_H]$, including $O_H = H_I$. The states $\lvert\phi(t)\rangle_I = U_{H_I(t)}(t, t_0)\lvert\phi(t_0)\rangle_S$ are time-dependent, with $H_I(t) = U^\dagger_{H_0}(t, t_0)H_I(t_0)U_{H_0}(t, t_0)$.

<table>
<tr><th colspan="2">Canonical quantisation steps</th></tr>
<tr><td>(P) Interaction picture</td><td>We consider a physical system described by a Lagrangian $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_I$ where $\mathcal{L}_0$ describes the free fields and $\mathcal{L}_I$ describes the interaction of the fields. Typically $\mathcal{L}_I$ is assumed to be smaller than $\mathcal{L}_0$ so that calculations can be made using perturbative series in $\mathcal{L}_I$.

We compute the general momentum field
$$\pi(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} = \frac{\partial \mathcal{L}_0}{\partial(\partial_0 \phi)}$$
as typically $\mathcal{L}_I$ does not depend on $\partial_0 \phi$, and define the Hamiltonian as
$$H = \int d^3\vec{x}\, [\pi(x)\partial_0\phi(x) - \mathcal{L}] = H_0 + H_I$$
where $H_0 = \int d^3\vec{x}\, [\pi(x)\partial_0\phi(x) - \mathcal{L}_0]$ and $H_I = -\int d^3\vec{x}\mathcal{L}_I$.

We use the interaction picture, with the basis states defined as eigenstates for the free-field Hamiltonian H_0. These basis states are used to define the incoming and outgoing particles at "asymptotic times", before and after the interactions occur (this is a debatable assumption as the interactions may not disappear at asymptotic times).
The free-field Hamiltonian H_0 gives also the time evolution of all operators, inlcluding ϕ and π.
We denote by $|0\rangle$ the vaccum state for the free-field theory H_0, i.e. $H_0 |0\rangle = 0$, $\langle 0|0\rangle = 1$.
We denote by $|\Omega\rangle$ the vaccum state for the full theory H, i.e. $H |\Omega\rangle = 0$, $\langle \Omega|\Omega\rangle = 1$.</td></tr>
<tr><td>(P) Canonical quantisation</td><td>To allow computation of H_0 and H_I, the fields $\phi(x)$ and $\pi(x)$ are quantized for the free field theory $\mathcal{L}_0$, H_0 i.e. promoted as fields of operators, where we impose the following constraints:

(1) The field $\phi(x)$ obeys the Euler-Lagrange equation obtained from the free-field Lagrangian $\mathcal{L}_0$ (contrary to path-integral formulation where the field has no such constraint):
$$\frac{\partial \mathcal{L}_0}{\partial \phi} - \frac{\partial}{\partial x^\mu}\frac{\partial \mathcal{L}_0}{\partial(\partial_\mu \phi)} = 0,$$
(2) The field $\phi(x)$ and $\pi(x)$ obey the commutation relations:

For fields describing Bose particles:
$[\phi_a(x), \pi_b(y)] = i\delta(x-y)\delta_{ab}$
$[\phi_a(x), \phi_b(y)] = 0, \quad [\pi_a(x), \pi_b(y)] = 0$
For fields describing Fermi particles:
$\{\phi_a(x), \pi_b(y)\} = i\delta(x-y)\delta_{ab}$
$\{\phi_a(x), \phi_b(y)\} = 0, \quad \{\pi_a(x), \pi_b(y)\} = 0$
where we made explicit the degrees of freedom for the fields $\phi = \{\phi_a\}_a$ and $\pi_a = \partial\mathcal{L}/\partial(\partial_0\phi_a)$.

(3) The operators are normal ordered, so that all operators creating particles are placed to the left compared to the operators annihilating the same particles ($(-1)\times$ applies when permuting fermion operators). A normal ordered product of operators $\phi(x_1)\cdots\phi(x_2)$ is denoted $:\phi(x_1)\cdots\phi(x_2):$</td></tr>
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<tr><td>(D) Scattering matrix</td><td>The scattering matrix S is the operator defined by
$$S = U_{H_I}(t_+, t_-) = T\left[\exp\left(-i\int_{t_-}^{t_+} H_I(s)ds\right)\right]$$
so that for the initial state $|i\rangle$ at asympotic time $t_- \to -\infty$ and the final state $|f\rangle$ at asymptotic time $t_+ \to +\infty$ (recall they are eigenstates of H_0), $\langle f|\, S\, |i\rangle$ is the probability amplitude to observe $|f\rangle$ starting from state $|i\rangle$.</td></tr>
<tr><td>(P) n-point Green function or correlation function</td><td>The n-point Green function or correlation function is defined as
$$G^{(n)}(x_1, \ldots, x_n) = \langle\Omega|\, T\left[\phi_H(x_1)\cdots\phi_H(x_n)\right] |\Omega\rangle$$
where the states and the field operators are in the Heisenberg picture for the full Hamiltonian H.
(Assuming that the spectrum for H gives an isolated minimum energy 0,) we can show that:
$$G^{(n)}(x_1, \ldots, x_n) = \frac{\langle 0|\, T\left[\phi_I(x_1)\cdots\phi_I(x_n)S\right] |0\rangle}{\langle 0|\, S\, |0\rangle}$$
where the numerator $\langle 0|\, S\, |0\rangle$ corresponds to vacuum bubbles which corresponds to particles creation/annihilation in the vaccum not connected to the ingoing/outgoing particles. The factoring on the right hand side says that we only need to consider processes which are connected to the ingoing/outgoing particles.</td></tr>
<tr><td colspan="2">(Proof) Using $S = T\left[\exp\left(-i\int_{t_-}^{t_+} H_I(s)ds\right)\right]$, the numerator reads $\langle 0|\, T\left[\phi_I(x_1)\cdots\phi_I(x_n)S\right] |0\rangle$ $= \langle 0|\, U_I(t_+, t_1)\phi_I(x_1)U_I(t_1, t_2)\phi_I(x_2)\cdots\phi_I(x_n)U_I(t_n, t_-)\, |0\rangle = \langle 0|\, U_I(t_+, t_0)\phi_H(x_1)\cdots\phi_H(x_n)U_I(t_0, t_-)\, |0\rangle$. Assume now that the spectrum of H has an isolated eigenvalue 0 (of eigenstate $|\Omega\rangle$) and a continuum spectrum $E > 0$. We have, for any state $|\psi\rangle$, $\langle\psi|\, U_I(t_0, t_-)\, |0\rangle = \langle\psi|\, U_I(t_0, t_-)[|\Omega\rangle\langle\Omega| + \int dE\, |E\rangle\langle E|]\, |0\rangle =$ $\langle\psi|\, U_I(t_0, t_-)\, |\Omega\rangle\langle\Omega|0\rangle + \int dE\, \langle\psi|\, U_I(t_0, t_-)\, |E\rangle\langle E|0\rangle$. The last term reads $\int dE\, \langle\psi|E\rangle\, e^{-iE(t_0 - t_-)}\langle E|0\rangle$ which converges to 0 as $(t_0 - t_-)$ tends to infinity using the Riemann Lebesgue theorem. We get $G^{(n)}(x_1, \ldots, x_n) = \frac{\langle 0|\Omega\rangle\langle\Omega|T[\phi_H(x_1)\cdots\phi_H(x_n)]|\Omega\rangle\langle\Omega|0\rangle}{\langle 0|\Omega\rangle\langle\Omega|\Omega\rangle\langle\Omega|0\rangle}$ which concludes the proof.</td></tr>
<tr><td>(P) How to get the S matrix from the correlation function
Lehmann-Symanzik-Zimmermann (LSZ) formula</td><td>First step is to perform the Fourier transform
$$\tilde{G}^{(n)}(p_1, \ldots, p_n) = \int\left[\prod_{i=1}^{n} d^4x_i e^{-ip_ix_i}\right] G^{(n)}(x_1, \ldots, x_n)$$
This however contains the propagators for the external legs ($i/(p^2 - m^2 + i\epsilon)$ for the scalar field for instance) so you need to cancel them by multiplying by their inverses. For a scalar field we get:
$$\langle p'_1, \ldots p'_{n'}|\, S - \mathbf{1}\, |p_1, \ldots, p_n\rangle = (-i)^{n+n'}$$
$$\times\prod_{i=1}^{n'}({p'_i}^2 - m^2)\prod_{j=1}^{n}(p_j^2 - m^2)\tilde{G}^{(n)}(-{p'}_1, \ldots, -p'_{n'}, p_1, \ldots, p_n)$$
Noting that the correlation functions are given with all momenta ingoing. The LSZ formula is valid provided that the field verifies (shift and multiply ϕ by constants if needed):
$$\langle\Omega|\, \phi_H(x)\, |\Omega\rangle = 0 \quad \text{and} \quad \langle k|\, \phi_H(x)\, |\Omega\rangle = e^{-ikx}$$
for any position x and one particle state $|k\rangle$ of momentum k.</td></tr>
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<tr><td colspan="2">(<u>Proof</u>) We have $\langle p_1', \dots p_{n'}' | \, S - \mathbf{1} \, | p_1, \dots, p_n \rangle = \langle f | i \rangle$ with $|i\rangle = \lim_{t \to -\infty} a_1^\dagger(t) \, |\Omega\rangle$ and $|f\rangle = \lim_{t \to +\infty} a_{1'}^\dagger(t) \, |\Omega\rangle$ for one-particle states for instance, provided $\langle \Omega | \, \phi(x) \, |\Omega\rangle = 0$ (so that $a^\dagger$ does not create a multi-particle state but a purely single particle state) and provided $\langle k | \, \phi(x) \, |\Omega\rangle = e^{-ikx}$ (so that such single-particle state is normalized). We have defined $a_1^\dagger(t) = \int d\vec{k} f_1(\vec{k}) a^\dagger(\vec{k})$ where $f_1(\vec{k}) \propto \exp[-(\vec{k} - \vec{k_1})^2/4\sigma^2]$ describes a wavepacket around momentum $\vec{k}_1$. We have $a^\dagger(\vec{k}) = -i \int d\vec{x} (e^{ikx} \partial_0 \phi(x) - ik_0 e^{ikx} \phi(x))$ and $a_1^\dagger(+\infty) - a_1^\dagger(-\infty) = \int_{-\infty}^{\infty} dt \partial_0 a_1^\dagger(t) = -i \int d\vec{k} f_1(\vec{k}) \int d^4x e^{ikx} (-\partial^2 + m^2) \phi(x)$. Applying the time ordering operator we have $a_{1'}(-\infty) \, |\Omega\rangle = 0$ and $\langle \Omega | \, a_1(+\infty) = 0$. The only terms remaining are the one given by the LSZ formula, for $\sigma \to 0$ which gives Diract functions $\delta(\vec{k} - \vec{k}_1)$.</td></tr>
<tr><td>(D) Feynman propagator and contraction</td><td>The Feynman propagator Δ_F is defined by
$$\Delta_F(x-y) = \langle 0 | \, T \left[\phi(x) \phi(y) \right] |0\rangle$$
For a product of fields
$$\cdots \phi(x_{i+1}) \phi(x_i) \phi(x_{i-1}) \cdots \phi(x_{j+1}) \phi(x_j) \phi(x_{j-1})$$
contracting a pair of fields $\phi(x_i)$ and $\phi(x_j)$ consists in replacing these two operators with the Feynman propagator $\Delta_F(x_i - x_j)$.
$$\cdots \phi(x_{i+1}) \overbrace{\phi(x_i) \phi(x_{i-1}) \cdots \phi(x_{j+1}) \phi(x_j)} \phi(x_{j-1}) = \\ = \Delta_F(x_i - x_j) \cdots \phi(x_{i+1}) \phi(x_{i-1}) \cdots \phi(x_{j+1}) \phi(x_{j-1})$$</td></tr>
<tr><td>(P) Wick's theorem</td><td>For a product of fields $\phi(x_1) \cdots \phi(x_n)$,
$$T[\phi(x_1) \cdots \phi(x_n)] =: \phi(x_1) \cdots \phi(x_n) : + : \text{all contractions} :$$
Example for $\phi_1 \phi_2 \phi_3 \phi_4$ where $\phi_i = \phi(x_i)$
$$T[\phi_1\phi_2\phi_3\phi_4] =: \phi_1\phi_2\phi_3\phi_4 : + \Delta_F(x_1 - x_2) : \phi_3\phi_4 : \\ + \Delta_F(x_1 - x_3) : \phi_2\phi_4 : + \text{(four similar terms)} \\ + \Delta_F(x_1 - x_2)\Delta_F(x_3 - x_4) + \Delta_F(x_1 - x_3)\Delta_F(x_2 - x_4) \\ + \Delta_F(x_1 - x_4)\Delta_F(x_2 - x_3).$$
This allows the computation of the Green function or the Scattering matrix with perturbative expansion in H_I.</td></tr>
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Canonical quantisation for real scalar fields	
(D) **Real scalar field** **Klein Gordon equation**	Consider a one-dimensional real field $\phi(x)$ with the free field Lagrangian: $\mathcal{L}_0 = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2\phi^2$ The Euler-Lagrange equation gives $\partial_\mu \partial^\mu \phi + m^2 \phi = 0$ called the **Klein-Gordon** equation. The momentum is $\pi = \partial_0 \phi$ The Hamiltonian is $H = \int d^3x \left[\frac{1}{2}\pi^2 + \frac{1}{2}\partial_k \partial^k \phi + \frac{1}{2}m^2\phi^2\right]$
(P) Fourier transform Creation/annihilation operators	Consider the generic Fourier transform for $\phi(x)$: $\phi(x) = \int \frac{d^4p}{(2\pi)^4}\phi(p)e^{-ipx}$ Applying the Euler-Lagrange equation we get the constraint on p_0: $p_0 = \pm E_{\vec{p}}$ where $E_{\vec{p}} = \sqrt{m^2 + \vec{p}^2}$ $\phi(x) = \int \frac{d^4p}{(2\pi)^4}\phi(p)e^{-ipx}\delta(p^2 - m^2)\theta(p_0) + \int \frac{d^4p}{(2\pi)^4}\phi(p)e^{-ipx}\delta(p^2 - m^2)\theta(-p_0)$ By changing the variable $p \mapsto -p$ in the second term and by noting that ϕ is real and labelling $a_{\vec{p}} = \phi(p)/\sqrt{2E_{\vec{p}}}$, $a^\dagger_{\vec{p}} = \phi(-p)/\sqrt{2E_{\vec{p}}} = \phi(p)^\dagger/\sqrt{2E_{\vec{p}}}$ (with $p_0 = E_p$) we get: $\phi(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\vec{p}}}}\left[a_{\vec{p}}e^{-ipx} + a^\dagger_{\vec{p}}e^{+ipx}\right]$ Using $\pi(x) = \partial_0\phi(x)$ we have $\pi(x) = \int \frac{d^3p}{(2\pi)^3}(-i)\sqrt{\frac{E_{\vec{p}}}{2}}\left[a_{\vec{p}}e^{-ipx} - a^\dagger_{\vec{p}}e^{+ipx}\right]$ and we see that the Bose particle commutation relation $[\phi(x), \pi(y)] = i\delta(x-y)$, $[\phi(x), \phi(y)] = [\pi(x), \pi(y)] = 0$ is satisfied if and only if $\left[a_{\vec{p}}, a^\dagger_{\vec{q}}\right] = (2\pi)^3\delta(\vec{p} - \vec{q})$ and $\left[a_{\vec{p}}, a_{\vec{q}}\right] = \left[a^\dagger_{\vec{p}}, a^\dagger_{\vec{q}}\right] = 0$
(P) Hamiltonian, Fock space	From π and ϕ we get the free field Hamiltonian H_0 after normal ordering $H_0 = \int \frac{d^3p}{(2\pi)^3}E_{\vec{p}}\, a^\dagger_{\vec{p}} a_{\vec{p}}$ and we find $\left[H_0, a^\dagger_{\vec{p}}\right] = E_{\vec{p}}a^\dagger_{\vec{p}}$ and $\left[H_0, a_{\vec{p}}\right] = -E_{\vec{p}}a_{\vec{p}}$. If $\|\epsilon\rangle$ is an eignestate of H_0 for energy ϵ then $a_{\vec{p}}\|\epsilon\rangle$ is an eignestate of H_0 for energy $(\epsilon - E_{\vec{p}})$ and $a^\dagger_{\vec{p}}\|\epsilon\rangle$ is an eignestate of H_0 for energy $(\epsilon + E_{\vec{p}})$. Define the vacuum state $\|0\rangle$ for the free field as the eigenstate of H_0 for energy 0. Then $a_{\vec{p}}\|0\rangle = 0$ and we define the states $a^\dagger_{\vec{p_1}} \cdots a^\dagger_{\vec{p_n}}\|0\rangle = \|p_1, \ldots p_n\rangle$ as the state with Bose particles $p_1, \ldots, p_n$. The state is symmetric under permutation of particles as the operators $a^\dagger_{\vec{p}}$ are.

(P) Feynman propagator for real fields	The Feynman propagator for a real field reads $$\begin{aligned}\Delta_F(x-y) &= \langle 0\| T[\phi(x)\phi(y)] \|0\rangle \\ &= \int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_{\vec{p}}}e^{-ip\cdot(x-y)\theta(x^0-y^0)+ip\cdot(x-y)\theta(y^0-x^0)} \\ &= \int \frac{d^4p}{(2\pi)^4}\frac{ie^{-ip(x-y)}}{p^2-m^2+i\epsilon}\end{aligned}$$ where the $+i\epsilon$ called the "Feynman prescription" is here so that we select the correct pole in the contour integration depending on the sign of (x^0-y^0) when doing the contour integration in p_0. The Feynman propagator is a Green's function for the Klein-Gordon operator, i.e.: $$\left(\frac{\partial}{\partial x^\mu}\frac{\partial}{\partial x_\mu}+m^2\right)\Delta_F(x-y) = -i\delta(x-y)$$

Canonical quantisation for spinor fields	
(D) **Spinor fields**	The spinor field is a complex 4-dimensional vector field ψ, $\overline{\psi} = \psi^\dagger \gamma^0$, with the free field Lagrangian: $\mathcal{L}_0 = \overline{\psi}(x)(i\partial\!\!\!/ - m)\psi(x)$ where ψ and $\overline{\psi}$ should be considered as independent variables (or you need to consider the real part and the imaginary part as independent variables). The Euler-Lagrange equation applied on $\overline{\psi}$ gives $(i\partial\!\!\!/ - m)\psi(x) = 0$ Applying $(i\partial\!\!\!/+m)$ on the left we see that each of the 4 components of $\psi(x)$ obeys the Klein-Gordon equation $(\partial_\mu \partial^\mu + m^2)\psi_\mu(x) = 0$ The momentum is $\pi = i\psi^\dagger$ The Hamiltonian is $H_0 = \int d^3x \overline{\psi}(-i\gamma^k \partial_k + m)\psi \quad$ (caution sum over $k = 1, 2, 3$)
(P) Fourier transform Creation/annihilation operators	Consider the generic Fourier transform for $\psi(x)$, with a decomposition in a basis $\{w^s(p)\}_s$ for the four-vector space: $\psi(x) = \sum_s \int \frac{d^4p}{(2\pi)^4} \psi^s(p) w^s(p) e^{-ipx}$ as for the real field, applying the Klein-Gordon equation constrains p_0 to be $\pm E_{\vec{p}}$ (but without the complex conjugate condition as ψ is not a real field). Applying the Dirac equation to the $w^s(p)e^{\mp ipx}$ we get $\begin{pmatrix} \mp m & p_\mu \sigma^\mu \\ p_\mu \overline{\sigma}^\mu & \mp m \end{pmatrix} w^s(p) = 0 \quad$ i.e. $\quad w^s = \begin{pmatrix} \sqrt{p \cdot \sigma}\xi \\ \pm\sqrt{p \cdot \overline{\sigma}}\xi \end{pmatrix}$ where ξ is a 2-component spinor. We have only two basis vectors for ξ, say $\xi^{s=1} = (1, 0)$ and $\xi^{s=2} = (0, 1)$. Therefore we have only two possible four-vectors for $w^s(p)$ and we finally get: $\psi(x) = \sum_{s=1,2} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left[b^s_{\vec{p}} u^s(\vec{p}) e^{-ipx} + c^{s\dagger}_{\vec{p}} v^s(\vec{p}) e^{+ipx} \right]$ for some operators $b^s_{\vec{p}}$ and $c^s_{\vec{p}}$, where $u^s(\vec{p}) = (\sqrt{p \cdot \sigma}\xi^s, \sqrt{p \cdot \overline{\sigma}}\xi^s)$, $v^s(\vec{p}) = (\sqrt{p \cdot \sigma}\xi^s, -\sqrt{p \cdot \overline{\sigma}}\xi^s)$. We have the useful identities: $\sum_s u^s(\vec{p})\overline{u}^s(\vec{p}) = p\!\!\!/ + m$, and $\quad \sum_s v^s(\vec{p})\overline{v}^s(\vec{p}) = p\!\!\!/ - m$ We get $\pi(x)$ trivially using $\pi(x) = i\psi^\dagger(x)$, and we see that the Fermi particle commutation relation $\{\psi_r(x), \pi_s(x)\} = i\delta_{rs}\delta(x-y)$, $\{\psi_r(x), \psi_s(x)\} = \{\pi_r(x), \pi_s(x)\} = 0$ is satisfied if and only if $\left\{ b^r_{\vec{p}}, b^{s\dagger}_{\vec{q}} \right\} = (2\pi)^3 \delta^{rs} \delta(\vec{p} - \vec{q})$ $\left\{ c^r_{\vec{p}}, c^{s\dagger}_{\vec{q}} \right\} = (2\pi)^3 \delta^{rs} \delta(\vec{p} - \vec{q})$ and all other anti-commutators are zero (between b and c as well).

<table>
<tr><td>(P) Hamiltonian, Fock space</td><td>From π and ψ we get the free field Hamiltonian H_0 after normal ordering
$$H_0 = \int \frac{d^3p}{(2\pi)^3} E_{\vec{p}} \left[b^{s\dagger}_{\vec{p}} \, b^s_{\vec{p}} + c^{s\dagger}_{\vec{p}} \, c^s_{\vec{p}} \right]$$
and we find $\left[H_0, b^{s\dagger}_{\vec{p}}\right] = E_{\vec{p}} b^{s\dagger}_{\vec{p}}$ and $\left[H_0, b^s_{\vec{p}}\right] = -E_{\vec{p}} b^s_{\vec{p}}$, and likewise for $c^s_{\vec{p}}$ as we had for the real field.
As before we can create states with particles and antiparticles using the operators $b^{s\dagger}_{\vec{p}}$ and $c^{s\dagger}_{\vec{p}}$ but this time note that these operators anti-commute, i.e. for instance, $b^{s\dagger}_{\vec{p}} b^{r\dagger}_{\vec{q}} \left|0\right\rangle = -b^{r\dagger}_{\vec{q}} b^{s\dagger}_{\vec{p}} \left|0\right\rangle$ and in particular, $b^{s\dagger}_{\vec{p}} b^{s\dagger}_{\vec{p}} \left|0\right\rangle = 0$ leading to the Pauli exclusion principle.</td></tr>
<tr><td>(P) Feynman propagator for spinors</td><td>The Feynman propagator for the spinor field reads
$$\begin{aligned} S_F(x-y) &= \left\langle 0\right| T[\psi(x)\overline{\psi}(y)] \left|0\right\rangle \\ &= \begin{cases} \left\langle 0\right| \psi(x)\overline{\psi}(y) \left|0\right\rangle & x^0 > y^0 \\ -\left\langle 0\right| \overline{\psi}(y)\psi(x) \left|0\right\rangle & x^0 < y^0 \end{cases} \\ &= \int \frac{d^4p}{(2\pi)^4} i e^{-ip(x-y)} \frac{\not{p} + m}{p^2 - m^2 + i\epsilon} \end{aligned}$$
The Feynman propagator is a Green function for the Dirac operator, i.e.
$$(i\not{\partial}_x - m) S_F(x-y) = i\delta(x-y)$$</td></tr>
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Canonical quantisation for photons	
(D) **Electromagnetic field, Lorentz Gauge**	The Lagrangian of the free electromagnetic field is, adding the Lorentz gauge condition $\partial_\mu A^\mu = 0$ in the Lagrangian: $$\mathcal{L}_0 = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2}(\partial_\mu A^\mu)^2$$ where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and $A_\mu(x)$ is a real Lorentz vector field. The Euler-Lagrange equation gives $$\partial_\mu \partial^\mu A^\nu = 0$$ and the momentum is $$\pi^0 = \frac{\partial \mathcal{L}_0}{\partial(\partial_0 A_0)} = -\partial_\mu A^\mu, \quad \pi^k = \frac{\partial \mathcal{L}_0}{\partial(\partial_0 A_k)} = \partial^k A^0 - \partial_0 A^k$$
(P) Fourier transform	Consider the generic Fourier transform for A^μ which is a 4-vector: $$A_\mu(x) = \sum_{\lambda=0}^{3} \int \frac{d^4p}{(2\pi)^4} \epsilon_\mu^\lambda(\vec{p}) e^{-ipx}$$ where we choose ϵ_μ^0 to be timelike, ϵ_μ^1 and ϵ_μ^2 to be transverse to the momentum i.e. $\epsilon^1 \cdot p = \epsilon^2 \cdot p = 0$, and ϵ_μ^3 to be longitudinal, i.e. aligned to $\vec{p}$. Applying Euler-Lagrange, we get the constraint $p^0 = \pm|\vec{p}|$, and as for the real field $$A_\mu(x) = \sum_{\lambda=0}^{3} \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\|\vec{p}\|}} \epsilon_\mu^\lambda(\vec{p}) \left[a_{\vec{p}}^\lambda e^{-ipx} + a_{\vec{p}}^{\lambda\dagger} e^{+ipx}\right]$$ $$\pi_\mu(x) = \sum_{\lambda=0}^{3} \int \frac{d^3p}{(2\pi)^3} \sqrt{\frac{\|\vec{p}\|}{2}} (+i) \epsilon_\mu^\lambda(\vec{p}) \left[a_{\vec{p}}^\lambda e^{-ipx} - a_{\vec{p}}^{\lambda\dagger} e^{+ipx}\right]$$ The commutation relations $[A_\mu(x), A_\nu(y)] = [\pi_\mu(x), \pi_\nu(y)] = 0$ and $[A_\mu(x), \pi_\nu(y)] = i\eta_{\mu\nu}\delta(x-y)$ are satisfied if and only if $[a_{\vec{p}}^\alpha, a_{\vec{q}}^\beta] = [a_{\vec{p}}^{\alpha\dagger}, a_{\vec{q}}^{\beta\dagger}] = 0$ and $$\left[a_{\vec{p}}^\alpha, a_{\vec{q}}^{\beta\dagger}\right] = -\eta^{\alpha\beta}(2\pi)^3\delta(\vec{p}-\vec{q})$$
(P) Hamiltonian Gupta-Bleuler condition	The Hamiltonian reads $$H = \int \frac{d^3p}{(2\pi)^3} \|\vec{p}\| \left(\sum_{k=1}^{3} a_{\vec{p}}^{k\dagger} a_{\vec{p}}^k - a_{\vec{p}}^{0\dagger} a_{\vec{p}}^0\right)$$ which yields positive eignevalues provided that we follow the **Gupta-Bleuler** condition: the physical states are described by states of the sub-Hilbert space $\mathcal{H}_{phys}$ defined by $\langle\psi\| \partial_\mu A^\mu \|\phi\rangle = 0$ for any $\|\psi\rangle, \|\phi\rangle$ in $\mathcal{H}_{phys}$.
(P) Feynman propagator for the electromagnetic field	The Feynman propagator for the electromagnetic field reads $$D_{\mu\nu}(x-y) = \langle 0\| T[A_\mu(x)A_\nu(y)] \|0\rangle = \int \frac{d^4p}{(2\pi)^4} ie^{-ip(x-y)} \frac{-i\eta_{\mu\nu}}{p^2 + i\epsilon}$$ The Feynamn propagator is a Green function for the electromagnetic operator, i.e. $$\frac{\partial}{\partial x^\mu}\frac{\partial}{\partial x_\mu} D_{\alpha\beta}(x-y) = i\delta(x-y)\eta_{\alpha\beta}$$

Quantum Electrodynamics	
(P) The full Lagrangian for photons and charged Fermi particles	$$\begin{aligned}\mathcal{L} &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \overline{\psi}(i\not{D} - m)\psi \\ &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \overline{\psi}(i\not{\partial} - m)\psi - e\overline{\psi}\not{A}\psi \\ &= \mathcal{L}_0 + \mathcal{L}_I\end{aligned}$$ where $\mathcal{L}_0 = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \overline{\psi}(i\not{\partial} - m)\psi$ describes the free electromagnetic and spinor fields, $\mathcal{L}_I = -e\overline{\psi}\not{A}\psi = -e\overline{\psi}\gamma^\mu A_\mu\psi$ describes the interaction between the electromagnetic and the spinor fields ($e \simeq 0.3$), and $D_\mu\psi = \partial_\mu\psi + ieA_\mu\psi$ is the **covariant derivative**.
(P) Calculation of scattering matrix	We compute the probability amplitude $\langle f\| S \|i\rangle$ to go from an initial state $\|i\rangle = \sqrt{2E_{\vec{p}}}a_{\vec{p}}^\dagger \cdots \|0\rangle$ to a final state $\|f\rangle = \sqrt{2E_{\vec{q}}}b_{\vec{q}}^\dagger \cdots \|0\rangle$ where $a^\dagger$ and $b^\dagger$ are appropriate particle creation operators. We perform the computation perturbatively (noting that $H_I = -\int \mathcal{L}_I d^3x$): $$S = \mathbf{1} + iT\left[\int \mathcal{L}_I(x)dx\right] + \frac{i^2}{2!}T\left[\int \mathcal{L}_I(x_1)\mathcal{L}_I(x_2)dx_1dx_2\right] + \cdots$$ What we get are vacuum expectation values of operators such as $\langle 0\| T[A_\mu(x_1)\psi(x_2)\cdots] \|0\rangle$ which are simplified using Wick's theorem and expressed as integrals on products of Feynamn propagators (in case of doubt you can decompose into creation/annihilation operators and do all the calculations).
(P) Feynman rules	We can deduce the following rules to compute the scattering matrix: **(1)** Consider only connected and amputated diagrams (no bubble in the vacuum, no bubble on external legs) **(2)** to each incoming fermion or outgoing anti-fermion, with momentum $\vec{p}$ and spin r, associate a spinor $u^r(\vec{p})$ **(3)** to each outgoing fermion or incoming anti-fermion, with momentum $\vec{p}$ and spin r, associate a spinor $\overline{u}^r(\vec{p})$ **(4)** to each incoming or outgoing photon, associate a polarisation vector ϵ^μ **(5)** each fermion-photon vertex gets a factor $(-ie\gamma^\mu)$ **(6)** each internal fermion line gets the propagator factor $\dfrac{i(\not{p}+m)}{p^2 - m^2 + i\epsilon}$ **(7)** each internal photon line gets the propagator factor $\dfrac{(-i\eta_{\mu\nu})}{p^2 + i\epsilon}$ **(8)** Momentum conservation at each vertex **(9)** Integrate over undetermined loop momenta **(10)** Consider permutation of identical particles with the minus sign for fermions

(P) Example: electron electron scattering	$$\langle f\| (S-\mathbf{1}) \|i\rangle = i\mathcal{A}(2\pi)^4\delta(p+q-p'-q')$$ with $$\mathcal{A} = (-ie)^2\Bigg(+ \frac{\left[\overline{u}^{s'}(p')\gamma^\mu u^s(p)\right](-i\eta_{\mu\nu})\left[\overline{u}^{r'}(q')\gamma^\nu u^r(q)\right]}{(p'-p)^2} - \frac{\left[\overline{u}^{s'}(p')\gamma^\mu u^r(q)\right](-i\eta_{\mu\nu})\left[\overline{u}^{r'}(q')\gamma^\nu u^s(p)\right]}{(p-q')^2}\Bigg)$$
(P) Example: electron positron scattering Bhabha scattering	$$\langle f\| (S-\mathbf{1}) \|i\rangle = i\mathcal{A}(2\pi)^4\delta(p+q-p'-q')$$ with $$\mathcal{A} = (-ie)^2\Bigg(- \frac{\left[\overline{u}^{s'}(p')\gamma^\mu u^s(p)\right](-i\eta_{\mu\nu})\left[\overline{v}^{r}(q)\gamma^\nu v^{r'}(q')\right]}{(p-p')^2} + \frac{\left[\overline{v}^{r}(q)\gamma^\mu u^s(p)\right](-i\eta_{\mu\nu})\left[\overline{u}^{s'}(p')\gamma^\nu v^{r'}(q')\right]}{(p+q)^2}\Bigg)$$ The (-1) of the first term is due to the fact that the positron-positron trajectory is obtained from the permutation in the time direction of an electron-electron trajectory in the electron-electron scattering.

(P) Example: electron-positron annihilation	$\langle f\vert\,(S-\mathbf{1})\,\vert i\rangle = i\mathcal{A}(2\pi)^4\delta(p+q-p'-q')$ with $\mathcal{A} = i(-ie)^2\overline{v}^r(q)\left(+ \frac{\gamma_\mu(\not{p}-\not{p}'+m)\gamma_\nu}{(p-p')^2-m^2} + \frac{\gamma_\nu(\not{p}-\not{q}'+m)\gamma_\mu}{(p-q')^2-m^2}\right)u^s(p)\epsilon_1^\nu(p')\epsilon_2^\mu(q')$